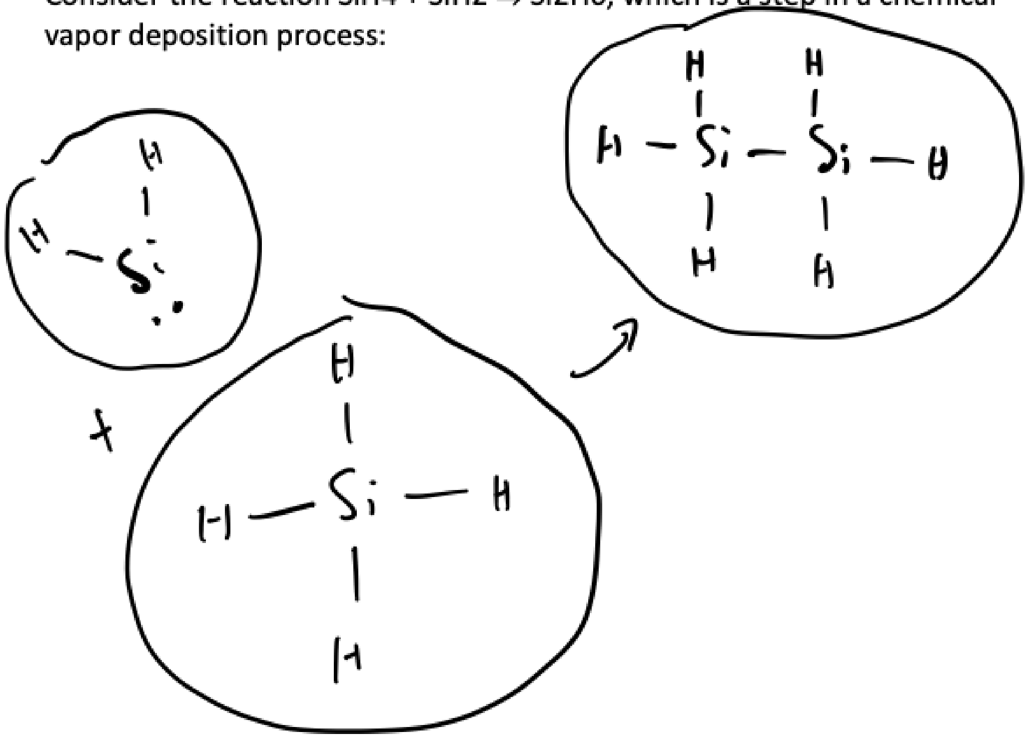
29 Reactions at the Molecular Level

Notes from https://sites.engineering.ucsb.edu/~jbraw/chemreacfun/ch2/slides-stoi-2up.pdf

Consider the reaction SiH4 + SiH2 ⇋ Si2H6, which is a step in a chemical vapor deposition process:



The reaction *extent* ξ keeps track of the number of times this reaction event occurs, i.e., it counts

the number of times an SiH_4 molecule combines with an SiH_2 molecule and forms a Si_2H_6 molecule

The reaction *rate* is then

$$r = \frac{\Delta \xi}{\Delta t\, V}$$ where Δt is "small" (10^{-13}-10^{-12} s)

Where V is large enough to comprise multiple molecules but small enough that it can be assumed to be well-mixed

If $\Delta\xi$ is positive:

reaction proceeds in the forward direction

If $\Delta\xi$ is negative:

reaction proceeds in the reverse direction

If the reaction is in equilibrium: $\Delta\xi = 0$, equal number of forward events & reverse events

Note that for this reaction, it is easy to "link" the rate of reaction to the rate of generation or consumption of one of the reactant or product species, since

$$r = -r_{SiH_4} = -r_{SiH_2} = r_{SiH_6}$$

We can thus measure the reaction rate based on the change in species concentrations, i.e., $\frac{1}{V}\frac{d\xi}{dt} \propto \frac{dC_j}{dt}$

However, for most reactions, it is not possible to measure the reaction rate. We hence must measure the rates of generation or production of individual species. This is why rates are often written in terms of specific reactant or product species

An example of such a reaction is 2NO + O2 --> 2NO2, where

$$r = -r_{NO} = -2r_{O_2} = r_{NO_2}$$

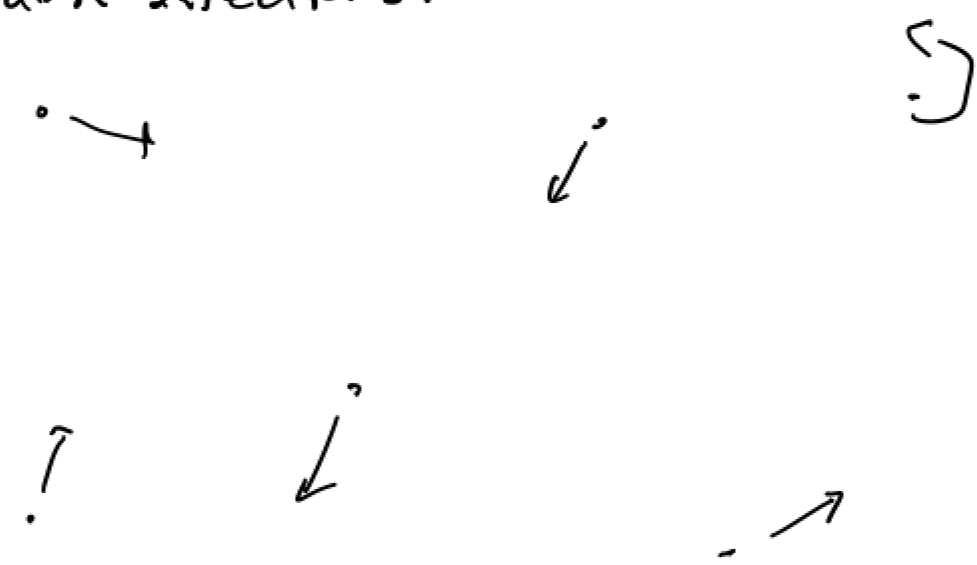
Consider the reaction 2NO + O2 --> 2NO2. Is it likely that this reaction could occur as a molecular event? Why or why not?

Let's answer this question by considering what must happen on the molecular level for the reaction to occur:

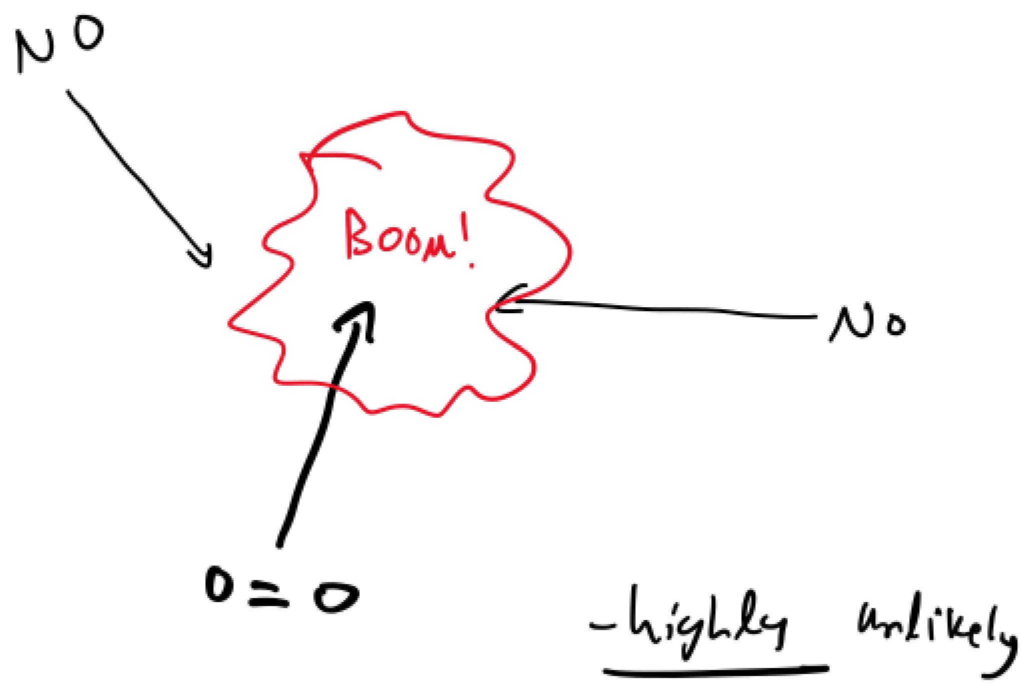
- O=O bond must break
- two O-NO bonds must form

Consider a homogeneous gas phase reaction. The gas molecules:

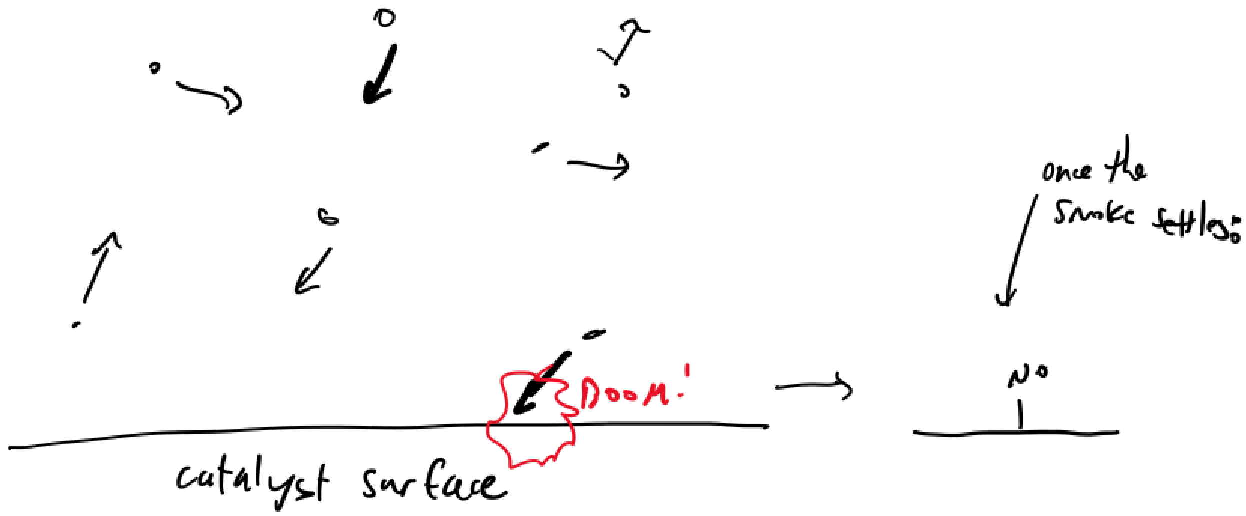
are moving at fast speeds in somewhat random directions.



For the reaction 2NO + O2 --> 2NO2 to happen as one molecular event:



Consider a gas phase reaction occurring over a catalyst surface:



For the reaction 2NO + O2 --> 2NO2 to happen as one molecular event:

Still very unlikely

So, to answer our question, no, this reaction is not likely to happen as a molecular event because

the likelyhood of 3 molecules colliding, let alone at an orientation that would favor reaction, is TINY.

So, we've looked at two reactions so far, SiH4 + SiH2 ⇋ Si2H6 and 2NO + O2 --> 2NO2. What if I told you that the reaction rates for these reactions were: $r_{Si_2H_6} = k_{Si_2H_6} C_{SiH_4} C_{SiH_2}$ and $r_{NO_2} = k_{NO_2} C_{NO}^2 C_{O_2}$ but that the first reaction is elementary <u>but</u> the second one <u>is not</u> 🤯

Reactions tend to convey two different meanings:

1) the change in the relative amounts of species that is observed when a reaction proceeds

2) The actual molecular events that happen as the reaction proceeds

The first is a stoichiometry statement the second is an elementary reaction

An elementary reaction is a change from reactants to products that proceeds without the formation of intermediate species

Elementary reactions are usually unimolecular or bimolecular because

the probability of species coming into contact decreases as the number of species increases

If the rate of a non-elementary reaction happens to exhibit elementary kinetics, it is

purely coincidental

I.e., the observed rate of 2NO + O2 --> 2NO2 is $r_{NO_2} = k_{NO_2} C_{NO}^2 C_{O_2}$, this means that the

reaction 2NO + O2 --> 2NO2 exhibits elementary kinetics
however, the

reaction $2NO + O_2 \rightarrow 2NO_2$ is <u>not an elementary reaction</u> since

it cannot proceed as a single molecular event

Rather, this reaction likely occurs via a reaction mechanism

which is comprised of multiple elementary steps

For example, consider the case where $2NO + O_2 \rightarrow 2NO_2$ occurs over a catalyst surface. One possible mechanism is:

1. $NO\ (g) + * \leftrightharpoons NO*$
1. $O_2\ (g) + * \leftrightharpoons O_2*$
1. $O_2* + * \leftrightharpoons 2O*$
1. $NO* + O* \leftrightharpoons NO_2* + *$
1. $NO_2* \leftrightharpoons NO_2\ (g) + *$

* = open catalyst site

NO* = NO bound to catalyst

*:

NO*:

catalyst surface

(b) the overall reaction 2NO + O2 --> 2NO2 is a linear combination

of reactions 1-5. The overall stoichiometry is reproduced by adding

2 × rxn 1 + rxn 2 + rxn 3 + 2 × rxn 4 + 2 × rxn 5.

Turning this into a vector:

$$s = \begin{array}{c} \begin{array}{ccccc} rxn1 & rxn2 & rxn3 & rxn4 & rxn5 \end{array} \\ \begin{bmatrix} 2 & 1 & 1 & 2 & 2 \end{bmatrix} \end{array}$$

Writing the stoichiometry matrix for the reaction mechanism:

$$\nu = \begin{array}{c} \begin{array}{cccccccc} NO & NO* & O_2 & O_2* & O* & NO_2 & NO_2* & * \end{array} \\ \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 2 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{bmatrix} \end{array}$$

We can then multiply s by ν to obtain

$$\begin{array}{c} \begin{array}{cccccccc} NO & NO* & O_2 & O_2* & O* & NO_2 & NO_2* & * \end{array} \\ \begin{bmatrix} -2 & 0 & -1 & 0 & 0 & 2 & 0 & 0 \end{bmatrix} \end{array}$$

which is the overall stoichiometry statement for the reaction, i.e.,

$$2NO + O_2 \rightarrow 2NO_2$$